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The Relationship Between Civil Pilots' Resilience,  
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**Abstract**

Resilience refers to individual's extant thinking process and behavior of choice which may protect the individual from suffering of robust stress or tragic events and smoothly get back to his/her life tracks. Resilience, thus, may be regarded as coping skills that the individual being equipped with in work organizations to improve his/her work performances from an occupational psychology perspective. That is, resilience may be defined, in part, as developing a constellation of traits to do individual's job well. For employees in high-stress and -risk aviation industry, it is of specialty crucial to assess the relationship between their resilience (i.e., stress coping skills), psychological well-being, and work performance. As such, individualized resilience-boosting program may be developed per the thorough understanding of this relationship. In the present study, 191 civil pilots were invited to participate in our questionnaire-filling survey. Based on factor analysis methods, six top-ranking factors encompassing 1) inner strength resource, 2) inner positive energy, 3) inspired growth, 4) solution by help, 5) interpersonal resource, and 6) spiritual religion were obtained for constructing our resilience scale. A total of three factors, including 1) work engagement, 2) promotion, and 3) good-teams, were used for establishing our work performance scale. Likewise, five factors consisting of 1) positive emotions, 2) commitment, 3) good relationships, 4) life meaning, and 5) achievement were employed to serve as our psychological well-being scale. The reliability tests revealed by Cronbach's alpha for the resilience, the psychological well-being, and the work performance scales ranged from .93 to .73. Using the correlation analysis methods, participants' resilience scores were positively associated with their psychological well-being ( $r=.85$ ,  $p<.001$ ), and work performance ( $r=.71$ ,  $p<.001$ ) scores. Moreover, each factor (aspect) of the resilience was correlated with each factor (aspect) in the psychological well-being and work performance scale in a positive manner (all  $ps<.001$ ). Furthermore, pilots' age seems to affect their two aspects, inner positive energy and spiritual religion, of resilience with paradoxical increments of these two factor scores in the eldest (age  $\geq 60$ ) group. Surprisingly, pilots with college degree had greater scores in inner positive energy and inspired growth of resilience as compared to pilots with advanced degree. Self-motivated ab initio group had greatest solution by help scores, while military ab initio group exhibited greatest spiritual religion scores in the three groups of ab initio training backgrounds. These results, taken together, prompt us to draw a tentative conclusion that our resilience scale may serve as a sensitive and reliable tool for not only providing civil aviators a convenient self-appraisal battery but concomitantly suggesting his/her mental health and well-being.

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## 1. Introduction

Civil aviation pilots are well-trained flight professionals. However, these pilots are susceptible to cognitive and behavioral impairments such as inability to concentrate and slow reactions under extreme fatigue and/or stress conditions (Cherng, et al., 2011; Cherng, et al., 2012; Sloan & Cooper, 1986). Workplace stress has been thought to negatively affect job performance and even more so to cause job dissatisfaction and/or burnout (Bharathi & Gupta, 2017). In the aviation industry, cognitive and behavioral impairments resulting from the stress of pilots are more likely to associate risks and accidents. Therefore, the stress control and response of civil aviation pilots has always been an important research topic. Failure of pilots to deal with stress appropriately is often identified as a significant cause of flight accidents (Alkov, et al., 1985; Caldwell, 2012; Hubbard, 2006; Leonhardt, 2016; Van Yperen & Hagedoorn, 2003). In contrast, appropriate stress coping skills have been regarded as trainable abilities to avoid or reduce the likelihood of aviation accidents. Accordingly, developing and self-assessing the positive stress coping ability, i.e., resilience (Britt et al., 2016), for pilots may be a feasible and convenient way to provide civil aviation pilots with a daily reflection of self-evaluation of resilience (present stress and coping results) and coping skill practice. A self-monitored resilience may be salutary and informative to provide a clearer-than-ever picture of psychological support and resources needed by civil pilots. Likewise, such practice may further maintain the physical and mental health of civil pilots, and thus ensure the flight safety of passengers.

### 1.1 Implications of resilience

Resilience refers to an individual's ability to face setbacks and challenges in life, to overcome obstacles, and to grow and empower in adversity (Walsh, 2003). Henry (1999) defines resilience as the ability of an individual to successfully cope, adjust, function positively and solve problems in situations of high risk, chronic stress or long-term trauma. That is, people with resilience are able to gradually recover their health from illness, depression or adversity (Abrams, 2001; Walsh, 2003). Anderson (1997) argues that resilience includes social, behavioral, and career capabilities that enable individuals to thrive in the face of poverty, interpersonal conflict, physical or psychological illness, and even harsh situations of childhood abuse or neglect. Rutter (1990; 2003) had shown that even though children experience many risk factors or difficult situations (such as parental divorce, poverty, abuse, etc.), more than half of those children may overcome these difficulties when they grow up, and become emotionally normal and socially well-developed adults. These observations inform the power of resilience.

Resilience can be subdivided into: health resilience (Sanders, et al, 2008), psychological resilience (Boardman, et al, 2008), emotional resilience (Chow, et al, 2007) and cognitive resilience (Rossi, et al, 2007). Resilience in health is primarily focused on the individual's recovery from physical pain, loss of function, and physical difficulty. Psychological resilience has dealt with the positive influence an individual has in facing various difficulties, disasters, and losses. For example, when acknowledging disasters such as air crashes and gas explosions, people are prone to fear, or beholding suicide incidents on campus or in flight, such as Malaysia Flight 370 on March 8, 2014 (Gladstone, 2016), as well as Germanwings Flight 9525 on March 24, 2015 (DeHoff, & Cusick, 2018), people are prone to fear of losing a loved one and a sense of security in day-to-day operations. The ability to bounce back to a normal pace of life in the face of these psychological unease is psychological resilience. Emotional resilience emphasizes the adjustment of positive and negative emotions in the face of stress. For instance, the emotional state of being disturbed by an uninvited guest, and the loss of not receiving due blessings or gifts on special festivals can all be calmed down to a normal emotional state. Cognitive resilience is the ability of an individual to overcome difficulties such as negative thoughts or cognitive disability in the wake of various stressful emotions. For example, when an individual is under

high pressure, he/she may be temporarily unable to think, analyze, judge, make decisions, etc., to adapt and change his/her thinking (Cherng & Yu, 2015).

In sum, resilience is the ability of an individual to return to a normal life trajectory from a variety of stressful and disturbing situations, including physical, emotional, financial or interpersonal difficulties. Individuals can gradually develop and change the degree of resilience through their own experiences. It is a dynamic process, influenced by life experiences and challenges (Cherng & Yu, 2015). In workplace, resilience can be viewed as a skill that an individual acquires from the organization or develops a skill enhancing job performance (Nguyen et al., 2016; Robertson et al., 2015), or a trait development (Van Veelen et al., 2019), and can positively respond to challenges in workplace (Britt et al., 2016). In theory, age and work tenures are conducive to the accumulation and demonstration of resilience (Ang et al., 2018; Caldeira & Timmins, 2016; Douglas & Pittenger, 2020; Gifkins et al., 2017; Hart et al., 2014). In a cohort study of organizational businesses, resilience has been shown to help individuals rebound from adversity and failure to normal functioning (Britt et al., 2016; Luthans, 2002), promote resilience, well-being, and job performance (Nguyen et al., 2016; Shin et al., 2012; Tugade et al., 2004). Numerous studies have supported that resilience is positively related to job performance, work engagement and job satisfaction, and that resilience is negatively related to emotional exhaustion and burnout (Luthans, 2002; Shoss et al., 2018; Youssef & Luthans, 2007).

Individual resilience may vary according to the diverse social contexts in which they live (Pietrzak & Southwick, 2011). That is, individuals may demonstrate adaptive resilience in their workplace but not so in their daily personal life. Therefore, workplace resilience training is extremely important to ensure individual's adaptive performance in their workplace. The organization (and employers) should be able to provide employees with the necessary resilience training to combat various types of stress in their employees' daily work (Kuntz et al., 2016; Robertson et al., 2015; Smith & Emerson, 2017) and to reduce negative effects of stress impinging upon the employees (Britt & Jex, 2015). Resilience training is currently used in medical teams (Mealer & Jones, 2014), military units (Crane & Boga, 2017), aviation units (Dekker et al., 2016; Douglas & Pittenger, 2020), and various other workplace use (Kossek & Perrigino, 2016).

To date, the development of assessment and measurement tools for workplace resilience have become an important part of current workplace practices (Kuntz et al., 2016; Rees et al., 2015). While there are many assessment tools to unveil the relationship between pilot personality, their job performance (Carretta et al., 2014; Goeters et al., 1993; Hormann & Maschke, 1996), and their physical and mental health (Cherng, Sher, & Chu, 2020; Cherng, 2018), to develop a sensitive, reliable and convenient pilot self-performing resilience kit is an unmet need in the context of intervention basis for civil pilots' selection, training, mental health promotion and mission-performing quality. The development of assessment tools for the resilience of civil aviation pilots is one of the most important measurement practice needed today (Douglas & Pittenger, 2020).

## ***1.2 Measurement of resilience***

Douglas and Pittenger (2020) have used The Connor-Davidson resilience scale (CD-RISC) (Connor & Davidson, 2003) to explore the resilience of female pilots using a 25-item measure of resilience derived from general population. Factor analysis has shown that a total of five factors are identified from the data obtained by the original CD-RISC in ordinary people. Factor 1: the individual's ability and high standards-related; Factor 2: the individual's tolerance for coping with stress-related; Factor 3: positive acceptance of change and safe interpersonal relationships-related; Factor 4: sense of control-related; Factor 5: the influence of the spiritual level-related. However, according to the factor analysis on the sample of those female pilots (Douglas & Pittenger, 2020), 6 items are removed due to their minor relevance. Following a second-wave factor analysis of the remaining 19 items, a total of two factors were obtained. Factor 1 has 10 items, mainly related to ability; Factor 2 has 9 items, mainly related to adaptation and persistence. They have compared the scores of the resilience scale of female pilots with those of other domestic cohorts and found that female pilots score significantly lower than other cohorts. In addition to highlighting a possibility that female pilots in the flight group need training in resilience and skill improvement in their workplace, these results are suggesting a lack of accuracy and applicability of such modified measurement tools. An urgent need is eminent to develop a sensitive and reliable resilience measurement tool suitable for specific workplaces, especially in the high-stress and high-risk aviation industry workplaces.

Cherng and Yu (2015) proposed that there are four main factors of resilience:

#### A. Inner strength resources:

- (1) Self-efficacy -- the individual's sense of self-control over one's own abilities, seeking external resources and advice to face problems, role-playing in problem-solving to obtain model learning from others, and problem-solving physical and psychological feedback and feelings.
- (2) Self-esteem--feeling the value of one's life and being able to "love" oneself with a positive and self-accepting attitude.
- (3) Determination -- when faced with problems or difficulties, individuals can have the ability and confidence to respond, focus, and seek solutions to remove obstacles.

These three inner strength resources are all related to the individual's problem solving ability, and also are similar to the hardy personality proposed by Kobasa (1982).

- B. Inner positive energy, including: optimism, positive attitude, and maintaining expectations and perceptions of things. Intuitively, facing problems with the positive energy of optimism, cheerfulness and humor can often resolve dilemmas.
- C. Interpersonal relationships, including interactions with family, friends, peers, and other acquaintances in social networks in an attempt to enhance mutual emotional, physical, and psychological assistance and social support, maintain a sense of social belonging, and benefit self-esteem and Self-efficacy needs satisfaction (Resnick, 2011).
- D. Spiritual and religion, including self-awareness, creating new things, experiencing or learning a new life, feeling humor or being curious about things, which are all conducive to a person's peaceful relationship with others. The experience of new things consists of learning new languages, new musical instruments, trying new activities, games and artistic activities. Participation in religious activities is thought to assist people overcome the sense of helplessness. Group prayer or chanting render the originally helpless people to re-understand and interpret the situations, and then feel the support and companionship of others. As such, helpless people are able to face the challenges and difficulties by gripping a surge of courage and hope, not despair. All these activities obliged to religions are the likely sources of energy that provides individual resilience.

### ***1.3 Factors affecting resilience and the current study goals***

#### A. Psychological well-being

Seligman (2011) has proposed five components of psychological well-being: positive emotions, engagement/commitment, good relationships, life meaning, achievement. Positive emotions are the manifestation of the inner positive energy. Among them, engagement/commitment and achievement are the inner strength resources. Good relationships and meaning in life are both directly related to relationships and spiritual and religious influences. Although it has been shown in organization and health Psychology studies that resilience is associated with well-being (Nguyen et al., 2016; Shin et al., 2012), there is no empirical study of civil aviation pilots investigating the relationship between resilience and well-being. Thus, to provide empirical evidence that civil pilots' stress coping ability (i.e., resilience) may associate with their well-being is the first goal of this study.

#### B. Work performance

As shown in Cherng & Yu (2020), work engagement is important part of work performance. Work engagement refers to individuals possessing positive, executive, and three psychological characteristics: vigor, dedication, and absorption in their workplace (Schaufeli, et al., 2002). Vigor includes being energetic at work and having good resilience, willing to work hard, and willing to persevere to the end and stick to the post unremittingly even encountering work difficulties. Dedication refers to workers see their work as important, serious, motivating, proud and challenging. Absorption means that workers can enjoy themselves and focus on their work, so they can feel a sense of smoothness and accomplishment in their work (van den Heavel, et al., 2010). Many studies support that

people with high work engagement have higher creativity, more achievements, and are more willing to take more responsibility at work (Bakker, et al., 2004; Bakker & Demerouti, 2008; Bakker & Schaufeli, 2008; Bakker, et al., 2008; Cartwright & Holmes, 2006; Hakanen, et al., 2005; May, et al., 2004; Schaufeli & Salanova, 2007; Schaufeli, et al., 2002; Xanthopoulou, et al., 2009). Cherng & Shiu (2015; 2016a, 2016b) had developed a short 10-item scale to measure work performance, consisting of work engagement, promotion, and good-teams. It is useful especially in the aviation workplace.

Research in organizational business has shown that resilience is positively correlated with job performance, job engagement, and job satisfaction (Shoss et al., 2018; Youssef & Luthans, 2007). In the field of civil aviation, to rigorously explore the relationship between civil pilots' resilience and their work performance is another goal of this study. Resilience refers to individual's extant thinking process and behavior of choice which may protect such an individual from suffering of robust stress or tragic events and smoothly get back to his/her life tracks. Resilience, thus, may be regarded as coping skills that the individual obtains from work organizations to improve his/her work performances. That is, resilience may be defined, in part, as the development of idiosyncratic and generalized traits to do individual's job well. In high-stress and -risk aviation industry, it is of relevance and importance to assess the relationship between individual's resilience, psychological well-being, and work performance.

## **2. Methods**

### **2.1 Participants**

A total of 191 pilots from domestic companies, including China Airlines, EVA Air, etc. participated in the questionnaire survey of this study. Participants were invited to respond to three custom-made scales as they received a routine physical examination at the Civil Medical Center. Participants' completed scales were collected and buck mailed back to PI for coding, data analysis and de-identification (anonymity) purposes.

### **2.2 Flight resilience scales**

Based on the literature of resilience, as well as the CD-RISC scale structure of Connor & Davidson (2003), the flight resilience scales consisting of 20 items for the aviation workplace were designed. Per the results of factor analysis, six factors were obtained, which explained a total of 77.35% of the variances. Along with the meaning of the items, they are named as: inner strength resources ( $\alpha=.93$ ), inner positive energy ( $\alpha=.86$ ), inspired growth ( $\alpha=.81$ ), interpersonal resources ( $\alpha=.86$ ), solution by help ( $\alpha=.74$ ), spiritual religion (single question), respectively.

### **2.3 Psychological well-being scales**

According to Seligman (2011)'s five components of psychological well-being: positive emotions, commitment, good relationships, life meaning, and achievement, the psychological well-being scales were designed. Likewise, factor analysis was employed and five factors were obtained, which explained 79.07% of the variances. According to the meaning of the items, five factors were named as: positive emotion ( $\alpha=.88$ ), commitment ( $\alpha=.86$ ), interpersonal relationships ( $\alpha=.84$ ), life meaning ( $\alpha=.73$ ), achievement ( $\alpha=.84$ ), respectively.

### **2.4 Work performance scales**

The work performance scales compiled by Cherng (Cherng & Shiu, 2015; 2016a, 2016b) were used. Three factors were obtained per the results of factor analysis, which explained 68.91% of the variances. They are named according to the meaning of the items: work engagement ( $\alpha=.82$ ), promotion ( $\alpha=.75$ ), and good-teams ( $\alpha=.78$ ).

### **2.5 Data analysis**

The research data were coding and analyzed by SPSS 24.0, including descriptive statistics, t test, one-way ANOVA, correlation coefficient analysis, factor analysis and internal reliability test.

### 3. Results

#### 3.1 Correlations of civil pilots' resilience, work performance, and psychological well-being

Individual civil pilot's mean scores for the three scales were calculated. We noticed that pilots' resilience scores correlated positively with their work performance scores ( $r = .71, p < .001$ ). Moreover, pilots' resilience scores correlated positively with their psychological well-being scores ( $r = .85, p < .001$ ). Furthermore, pilots' work performance scores correlated positively with their psychological well-being scores ( $r = .76, p < .001$ ) (Table 1). These results, taken together, suggest that civil pilots having low resilience scores are prone to exhibit below-averaged work performance or psychological well-being. In contrast, civil pilots having high resilience scores are also inclined to self-report their above-average work performance and psychological well-being.

Table 1. Correlations among pilots' resilience, work performance, and psychological well-being

|                          | Work performance | Psychological well-being | Resilience |
|--------------------------|------------------|--------------------------|------------|
| Work performance         | 1                |                          |            |
| Psychological well-being | .76***           | 1                        |            |
| Resilience               | .71***           | .85***                   | 1          |

\*\*\*:  $P < .001$

#### 3.2 Correlations between psychological well-being and resilience aspects

Applying correlation analysis between 6 factors (inner strength resource, inner positive energy, inspired growth, solution by help, interpersonal resource, and spiritual religion) of the resilience scale and the 5 factors (positive emotions, engagement, good relationships with others, life meaning, and achievement) of psychological well-being scale, the results were shown in Table 2. All of the paired factors from two scales were positive correlated ( $ps < .001$ ). That is, civil pilots self-reporting good psychological well-being usually maintained their traits of resilience.

Table 2. The relationships between 5 factors of psychological well-being and 6 factors of resilience

|                          | positive emotions | commitment | good relations | life meanings | achievement |
|--------------------------|-------------------|------------|----------------|---------------|-------------|
| inner strength resources | .642***           | .731***    | .691***        | .721***       | .716***     |
| inner positive energy    | .615***           | .653***    | .551***        | .686***       | .704***     |
| inspired growth          | .599***           | .686***    | .662***        | .690***       | .765***     |
| interpersonal resources  | .506***           | .545***    | .637***        | .527***       | .566***     |
| solution by helps        | .628***           | .714***    | .670***        | .701***       | .700***     |
| spiritual religion       | .416***           | .369***    | .395***        | .365***       | .365***     |

\*\*\*:  $P < .001$

#### 3.3 Correlations between work performance and resilience

Applying correlation analysis for 6 factors (inner strength resource, inner positive energy, inspired growth, solution by help, interpersonal resource, and spiritual religion) of resilience scale and 3 factors (work engagement, promotion, and good-teams) of work performance scale, the results were summarized in Table 3. All of factors were positively correlated ( $ps < .001$ ), suggesting the more work engagement/performance a pilot is, the more resilient he/she is.

Table 3. The relationships between work performance and resilience

|                          | work engagement | promotion | good-teams |
|--------------------------|-----------------|-----------|------------|
| inner strength resources | .615***         | .442***   | .600***    |
| inner positive energy    | .620***         | .385***   | .491***    |
| inspired growth          | .576***         | .399***   | .586***    |
| interpersonal resources  | .498***         | .518***   | .570***    |
| solution by helps        | .602***         | .370***   | .574***    |
| spiritual religion       | .410***         | .292***   | .376***    |

\*\*\*:  $P < .001$

### 3.4 Pilots' demographical characteristics and resilience

The score differences in resilience scale per civil pilot's sex, age, education, marriage, training background, tenure, flight hours were compared and contrasted. Applying unpaired t-tests (group-group difference) or one-way ANOVAs (more than three groups), there were no significant differences on the 6 factors of resilience scale according to the civil pilots' sex, status of marriage, tenure, and flight hours. Only age (Figure 1), education (Figure 2), and flight training background (Figure 3) may cause intriguing but statistically significant differences on some factors of the resilience scale. As shown in Figure 1, pilots at age of 60 and older were found to have the highest scores on both factors of inner positive energy and spiritual religion among five age groups of pilots. Those age groups differences on these two factors were statistically significant ( $F=2.80$ ,  $p<.05$ ;  $F=2.46$ ,  $p<.05$ , respectively).

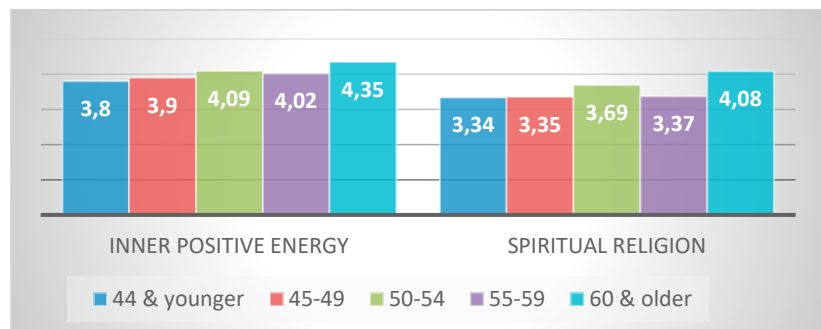


Figure 1. Age groups were different on their scores of inner positive energy and spiritual religion

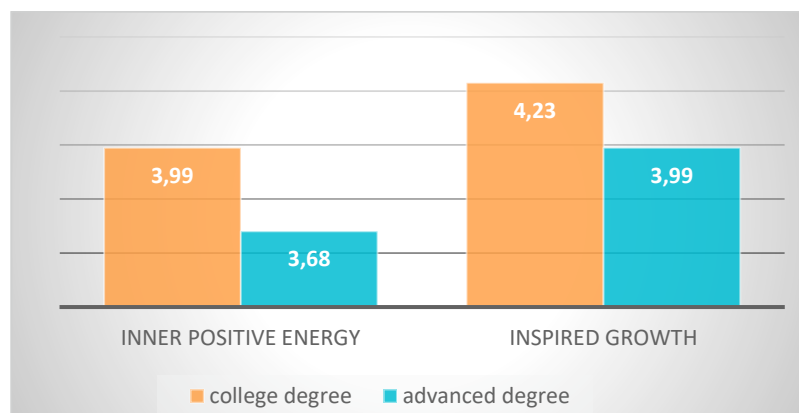


Figure 2. Two groups of education were different on their scores of inner positive energy and inspired growth

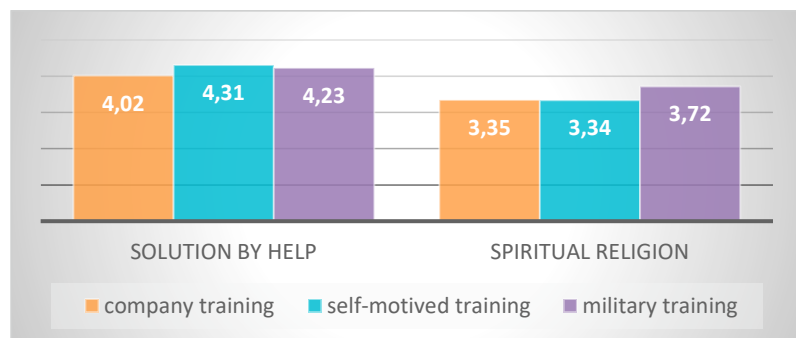


Figure 3. Three groups of ab initio training backgrounds were different on their scores of solution by help and spiritual religion

As shown in Figure 2, pilots with bachelor degree had the higher scores on both factors of inner positive energy and inspired growth than pilots with master degree did. These differences were statistically significant ( $F=6.68$ ,  $p<.05$ ;  $F=4.75$ ,  $p<.05$ , respectively). Finally, in Figure 3, pilots' initial flight training is self-motivated training had the highest scores on the factors of solution by help than the other two groups. However, pilots' initial flight training was motivated by military joining had the highest scores on the factor of spiritual religion. These differences were significant statistically ( $F=3.34$ ,  $p<.05$ ;  $F=3.08$ ,  $p<.05$ , respectively).

#### 4. Discussions

As shown in the current study results, pilots' resilience is indeed positively correlated with their psychological well-being and work performance. However, social approval and/or organization ethics may be paramount in accounting for pilots' self-report of high resilience, work performance and psychological well-being. Nonetheless, self-serving bias attribution and defense mechanisms, at least, play a role in predicting those pilots self-reporting their poor resilience along with poor self-evaluation on work performance and psychological well-being.

It is expected that pilots' resilient training is useful to keep pilots maintaining well-beings. Since the background of pilots, i.e., sex, marriage, tenure, and flight hours had no differences on their scores of resilience scale, it may display that resilience is the crucial character for civil aviation pilots. However, there are still some differences based on pilots' age, education, and flight training background. It may be useful to enhance their resilience according to those differences. The tentative conclusion as yet is that resilience scale seems to be a feasible tool for providing civil aviators a self-appraisal battery for indicating his/her mental health and well-being.

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#### References

- Abrams, M. S. (2001). Resilience in ambiguous loss. *American Journal of Psychotherapy*, 2, 283-291.
- Alkov, R.A., Gaynor, J.A. & Borowsky, M.S. (1985). Pilot Error as a Symptom of Inadequate Stress Coping. *Aviation, Space, and Environmental Medicine*, 56, 244-247.
- Anderson, K. M. (1997). Uncovering survival abilities in children who have been sexually abused: Families in society. *The Journal of Contemporary Human Services*, 78(6), 592-599.
- Ang, S. Y., Uthaman, T., Ayre, T. C., Mordiffi, S. Z., Ang, E., & Lopez, V. (2018). Association between demographics and resilience—a cross-sectional study among nurses in Singapore. *International Nursing Review*, 65(3), 459-466.
- Bakker, A.B., & Demerouti, E. (2008). Towards a model of work engagement. *Career Development International*, 13, 209-328.
- Bakker, A.B., Demerouti, E., & Verbeke, W. (2004). Using the job demands-resources model to predict burnout and performance. *Human Resource Management*, 43, 83-104.
- Bakker, A.B., & Schaufeli, W.B. (2008). Positive organizational behavior: Engaged employees in flourishing organizations. *Journal of Organizational Behavior*, 29, 147-154.
- Bakker, A.B., Schaufeli, W.B., Leiter, M.P., & Taris, T. (2008). Work engagement: An emerging concept in occupational health psychology. *Work & Stress*, 22, 187-200.
- Bharathi, T., & Gupta, K. S. (2017). Job stress and productivity: A conceptual framework. *International Journal of Emerging Research in Management & Technology*, 6(8), 393-398.
- Boardman, J.D., Blalock, C.L., & Button, T.M. (2008). Sex differences in the heritability of resilience. *Twin Research and Human Genetics*, 11 (1), 12-27.
- Britt, T. W., & Jex, S. M. (2015). *Thriving under stress: Harnessing demands in the workplace*. Oxford University Press.
- Britt, T. W., Shen, W., Sinclair, R. R., Grossman, M. R., & Klieger, D. M. (2016). How much do we really know about employee resilience? *Industrial and Organizational Psychology*, 9(2), 378-404.
- Caldeira, S., & Timmins, F. (2016). Resilience: Synthesis of concept analyses and contribution to nursing classifications. *International Nursing Review*, 63(2), 191-199.
- Caldwell, J. A. (2012). Crew schedules, sleep deprivation, and aviation performance. *Current Directions in Psychological Science*, 21(2), 85-89.
- Carretta, T., Teachout, M., Ree, M., Barto, E., King, R., & Michaels, C. (2014). Consistency of the relations of cognitive ability and personality traits to pilot training performance. *The International Journal of Aviation Psychology*, 24(4), 247-264.
- Cartwright, S., & Holmes, N. (2006). The meaning of work: The challenge of regaining employee engagement and reducing cynicism. *Human Resource Management Review*, 16, 199-208.
- Cherng, C.G. (2018). Exploring Successful Aging: Relationships between Age, Personality Functions and Successful Aging in Taiwanese Civil Pilots. In H. Becherstorfer, J. Lasry & M. Schwarz (Eds.), *Proceedings of the 33rd Conference CONNECTING PEOPLE, ORGANISATIONS AND TECHNOLOGY IN AVIATION*, pp 492-496. Dubrovnik, Croatia.
- Cherng, C-F, Sher, J-S., & Chu, H. (2020). The relationships of the big-five personality traits, sleeping qualities, and ages: An exploration study

- on civil pilots in Taiwan. The 14th European Academy of Occupational Health Psychology Conference (online conference), Cyprus: Nicosia.
- Cherng, C. G. & Shiu, J. (2015). Civil pilots' positive self-disclosure experiences in related to their job engagement and sleeping hours. 2015 Hong Kong International conference on education, psychology, and society (2015 HKICEPS), Kowloon, Hong Kong.
- Cherng, C. G. & Shiu, J. (2016a). Taiwan Airline pilots' secret keepings in related to their positive self-disclosure experiences and job engagements. The 32nd EAAP Conference: Thinking, High and Low: Cognition and Decision Making in Aviation, Cascais, Portugal.
- Cherng, C. G. & Shiu, J. (2016b). The positive impacts of civil aviators' self-disclosure experiences on their job engagement and job performance. The 12th European Academy of Occupational Health Psychology, Royal Olympic Hotel, Athens, Greece.
- Cherng, C. G., Shiu, J., & Wen, T-S. (2011). The impacts of Civil pilots' personality and stress on their flight performance. *Journal of Aviation Medicine & Sciences ROC*, 25(1), 3-18.
- Cherng, C. G., Shiu, J., & Wen, T-S. (2012). Civil pilots' stress and coping behaviors: A comparison between Taiwanese and non-Taiwanese aviators. In A. de Voogt & T. C.D'Oliveira (Eds), *Mechanisms in the Chain of Safety: Research and Operational Experiences in Aviation Psychology*, pp. 81-96. England: Ashgate.
- Cherng, C-F, Yu, L. (2015). *Gerontological Psychology*. Taipei: Yeh Yeh. (In Chinese)
- Cherng, C-F, Yu, L. (2020). *Aviation Psychology*, 2<sup>nd</sup> Edition. Taipei: Hung Yeh. (In Chinese)
- Chow, S.M., Hamagani, F., & Nesselroade, J.R. (2007). Age differences in dynamical emotion-cognition linkages. *Psychology and Aging*, 22(4), 765-780.
- Connor, K. M., & Davidson, J. R. (2003). Development of a new resilience scale: The Connor-Davidson resilience scale (CD-RISC). *Depression and Anxiety*, 18(2), 76–82.
- Crane, M., & Boga, D. (2017). A commentary: Rethinking approaches to resilience and mental health training. *Journal of Military and Veterans Health*, 25(1), 30.
- DeHoff, M. C., & Cusick, S. K. (2018). Mental Health in Commercial Aviation - Depression & Anxiety of Pilots. *International Journal of Aviation, Aeronautics, and Aerospace*, 5(5), Article 5.
- Dekker, S., Dahlström, N., van Winsen, R., & Nyce, J. M. (2016). Crew resilience and simulator training in aviation. In E. Hollnagel, C. P. Nemeth, & S. Dekker (Eds.), *Resilience engineering perspectives* (Vol. 1, pp. 133–140). CRC Press.
- Douglas, S., & Pittenger, L. M. (2020). Adversity in Aviation: Understanding Resilience in the Workplace for Female Pilots. *The International Journal of Aerospace Psychology*(Online), 89-103.
- Giffkins, J., Loudoun, R., & Johnston, A. (2017). Coping strategies and social support needs of experienced and inexperienced nurses performing shiftwork. *Journal of Advanced Nursing*, 73(12), 3079–3089.
- Gladstone, R. (2016). Pilot of vanished Malaysian flight had deviant route on his simulator, minister says. *The New York Times*.
- Goeters, K.-M., Timmermann, B., & Maschke, P. (1993). The construction of personality questionnaires for selection of aviation personnel. *The International Journal of Aviation Psychology*, 3(2), 123–141.
- Hakanen, J.J., Bakker, A.B., & Demerouti, E. (2005). How dentists cope with their job demands and stay engaged: The moderating role of job resources. *European Journal of Oral Sciences*, 113, 479-487.
- Hart, P. L., Brannan, J. D., & De Chesnay, M. (2014). Resilience in nurses: An integrative review. *Journal of Nursing Management*, 22(6), 720–734.
- Henry, D.L. (1999). Resilience in maltreated children: Implications for special needs adoptions. *Child Welfare*, 78(5), 519-540.
- Hormann, H.-J., & Maschke, P. (1996). On the relation between personality and job performance of airline pilots. *The International Journal of Aviation Psychology*, 6(2), 171–178.
- Hubbard, T. (2006). How we explain misfortune: Psychological implications for mental health in aviation professionals. In R. Bor & T. Hubbard (Eds), *Aviation Mental Health: Psychological implications for Air Transportation*, pp. 315-334. England: Ashgate.
- Kobasa, S. C., Maddi, S. R., & Kahn, S. (1982). Hardiness and Health: A prospective study. *Journal of Personality and Social Psychology*, 42, 168-177.
- Kossek, E. E., & Perrigino, M. B. (2016). Resilience: A review using a grounded integrated occupational approach. *The Academy of Management Annals*, 10(1), 729–797.
- Kuntz, J. R., Näswall, K., & Malinen, S. (2016). Resilient employees in resilient organizations: Flourishing beyond adversity. *Industrial and Organizational Psychology*, 9(2), 456–462.
- Leonhardt, J. (2016). Critical incident stress management in Air Traffic Control (ATC). In J. Leonhardt & J. Vogt (Eds.), *Critical Incident Stress Management in Aviation* (pp. 81–92). Routledge.
- Luthans, F. (2002). Positive organizational behavior: Developing and managing psychological strengths. *Academy of Management Executive*, 16(1), 57–72.
- May, D.R., Gilson, R.L., & Harter, L.M. (2004). The psychological conditions of meaningfulness, safety, and availability and the engagement of the human spirit at work. *Journal of Occupational and Organizational Psychology*, 77, 11-37.
- Mealer, M., & Jones, J. (2014). Methodological and ethical issues related to qualitative telephone interviews on sensitive topics. *Nurse Researcher*, 21(4), 4.
- Nguyen, Q., Kuntz, J. R., Näswall, K., & Malinen, S. (2016). Employee resilience and leadership styles: The moderating role of proactive personality and optimism. *New Zealand Journal of Psychology* (Online), 45(2), 13.
- Pietrzak, R., & Southwick, S. (2011). Psychological resilience in OEF-OIF veterans: Application of a novel classification approach and examination of demographic and psychosocial correlates. *Journal of Affective Disorders*, 133(3), 560–568.
- Rees, C. S., Breen, L. J., Cusack, L., & Hegney, D. (2015). Understanding individual resilience in the workplace: The international collaboration of workforce resilience model. *Frontiers in Psychology*, 6, 73.
- Resnick, B. (2011). Resilience in older adults. In M. L. Wykle, & S. H. Gueldner (Eds), *Aging well: Gerontological education for nurses and other health professionals*, pp 563-576. USA: Jones & Bartlett Learning, LTC.
- Robertson, I. T., Cooper, C. L., Sarkar, M., & Curran, T. (2015). Resilience training in the workplace from 2003 to 2014: A systematic review. *Journal of Occupational and Organizational Psychology*, 88(3), 533–562.
- Rossi, N.E., Bisconti, T.L., & Bergeman, C.S. (2007). The role of dispositional resilience in regaining life satisfaction after the loss of a spouse. *Death Studies*, 31 (10), 863-883.

- Rutter, M. (1990). Psychosocial resilience and protective mechanisms. In J. Roll, A. S. Masten, D. Cicchetti, K. Nuechterlein, & S. Weintraub (Eds.), *Risk and Protective Factors in the Development of Psychopathology* (pp.181-215). New York: Cambridge University Press.
- Rutter, M. (2003). Genetic influences on risk and protection: Implications for understanding resilience. In S. S. Luthar (Ed.), *Resilience and Vulnerability* (pp. 489-509). Cambridge, UK: Cambridge University Press.
- Sanders, A.E., Lim, S., & Sohn, W. (2008). Resilience to urban poverty : Theoretical and empirical considerations for population health. *American Journal of Public Health*, 98 (6) , 1101-1106.
- Schaufeli, W.B., & Salanova, M. (2007). Work engagement: An emerging psychological concept and its implications for organizations. In S.W. Gilliland, D.D., Steiner & D.P. Skarlicki (Eds.), *Research in social issues in management: Managing social and ethical issues in organizations*, Vol. 5, Greenwich, CT: Information Age Publishers.
- Schaufeli, W.B., Salanova, M., Gonzalez-Roma, V., & Bakker, A.B. (2002). The measurement of engagement and burnout: A two sampl confirmatory factor analytic approach. *Journal of Happiness Studies*, 3, 71-92.
- Seligman, M. (2011). *Flourish*. New York, NY: Free Press.
- Shin, J., Taylor, M. S., & Seo, M. G. (2012). Resources for change: The relationships of organizational inducements and psychological resilience to employees' attitudes and behaviors toward organizational change. *Academy of Management Journal*, 55(3), 727–748.
- Shoss, M. K., Jiang, L., & Probst, T. M. (2018). Bending without breaking: A two-study examination of employee resilience in the face of job insecurity. *Journal of Occupational Health Psychology*, 23(1), 112.
- Sloan, S. J. & Cooper, C. L. (1986). *Pilots Under Stress*. London, England: Routledge and Kegan Paul.
- Smith, K. J., & Emerson, D. J. (2017). An analysis of the relation between resilience and reduced audit quality within the role stress paradigm. *Advances in Accounting*, 37, 1–14.
- Tugade, M. M., Fredrickson, B. L., & Feldman Barrett, L. (2004). Psychological resilience and positive emotional granularity: Examining the benefits of positive emotions on coping and health. *Journal of Personality*, 72(6), 1161–1190.
- van den Heuvel, Machteld, Demerouti, E., Bakker, A. B., & Schaufeli, W. B. (2010). Personal resources and work engagement in the face of change. In J. Houdmont & Leka, S. (Eds.), *Contemporary Occupational Health Psychology: Global perspectives on research and practice*, pp 124-150. UK: Wiley-Blackwell.
- Van Veelen, R., Derks, B., & Endedijk, M. D. (2019). Double trouble: How being outnumbered and negatively stereotyped threatens career outcomes of women in STEM. *Frontiers in Psychology*, 10, 150.
- Van Yperen, N. W., & Hagedoorn, M. (2003). Do high job demands increase intrinsic motivation or fatigue or both? The role of job control and job social support. *Academy of Management Journal*, 46(3), 339–348.
- Walsh, F. (2003). Family resilience: A framework for clinical practice--Theory and practice. *Family Processes*, 42, 1-18.
- Xanthopoulou, D., Bakker, A.B., Demerouti, E., & Schaufeli, W.B. (2009). Reciprocal relationships between job resources, personal resources, and work engagement. *Journal of Vocational Behavior*, 74, 235-244.
- Youssef, C. M., & Luthans, F. (2007). Positive organizational behavior in the workplace: The impact of hope, optimism, and resilience. *Journal of Management*, 33(5), 774–800.